EWB COURSES | PROJECT REPORT

AWS VPC Peering

- Cross-VPC Private Communication
- Hands-on networking with Amazon VPC, EC2, route tables, security groups and a bastion host.
- VPC A → VPC PEERING → VPC B

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1. Project Overview

This project demonstrates private IPv4 communication between two Amazon Virtual Private Clouds (VPCs) using AWS VPC Peering. Route tables and security groups govern whether traffic can cross the peering connection. A bastion host provides SSH administration access to private workloads.

2. Problem Statement and Objectives

Resources in separate VPCs are isolated by default. The objective is to establish peering, configure forward and return routes, and validate connectivity between EC2 instances using private IPv4 addresses.

• Plan non-overlapping VPC CIDR ranges.

• Create public and private subnets with explicit route table associations.

• Configure VPC peering and routes on both sides.

• Use SSH ProxyJump through a bastion host.

• Test forward and reverse ICMP connectivity and document observed results.

3. Architecture and Resource Inventory

Resource	ID / CIDR / details
VPC A	vpc-0f04c9fb8933a150d — 10.10.0.0/16
VPC B	vpc-099cace4d50b0400f — 10.20.0.0/16
Public subnet A	subnet-0e4d721a1e4fbe561 — 10.10.1.0/24
Private subnet A	subnet-0f748cc67c6e3c6f7 — 10.10.2.0/24
Private subnet B	subnet-0f04d830bbbf8fac9 — 10.20.1.0/24
Bastion EC2	i-0eac0ba8e4d719410 — public IP earlier 13.201.47.28; later screenshot 13.201.0.96
App-A EC2	i-0bb30bd6ead4bf8a8 — earlier IP 10.10.1.155; another screenshot shows host 10.10.1.52
App-B EC2	i-0aed1caf9796939a4 — private IP 10.20.1.84 shown in terminal
Peering	pcx-075ba5989db37338f — reported Active

Important discrepancy: the planned App-A subnet is 10.10.2.0/24, but the observed App-A addresses 10.10.1.155 and 10.10.1.52 belong to 10.10.1.0/24. Verify the actual instance ID, private IP and subnet association before final submission.

4. Route Tables and Subnet Associations

Route table	Destination	Target	Status
ewb-rt-a-public	0.0.0.0/0	igw-00b5c8c0a8b4625cb	Active
ewb-rt-a-public	10.10.0.0/16	local	Active
ewb-rt-a-public	10.20.0.0/16	pcx-075ba5989db37338f	Active
ewb-rt-a-private	10.10.0.0/16	local	Active
ewb-rt-a-private	10.20.0.0/16	pcx-075ba5989db37338f	Active
ewb-rt-b-private	10.10.0.0/16	pcx-075ba5989db37338f	Active
ewb-rt-b-private	10.20.0.0/16	local	Active

Route table IDs

Name	ID
ewb-rt-a-public	rtb-08b22623ec522319c
ewb-rt-a-private	rtb-0deafce15e791e90f
ewb-rt-b-private	rtb-0d09e1b8f2f14319c

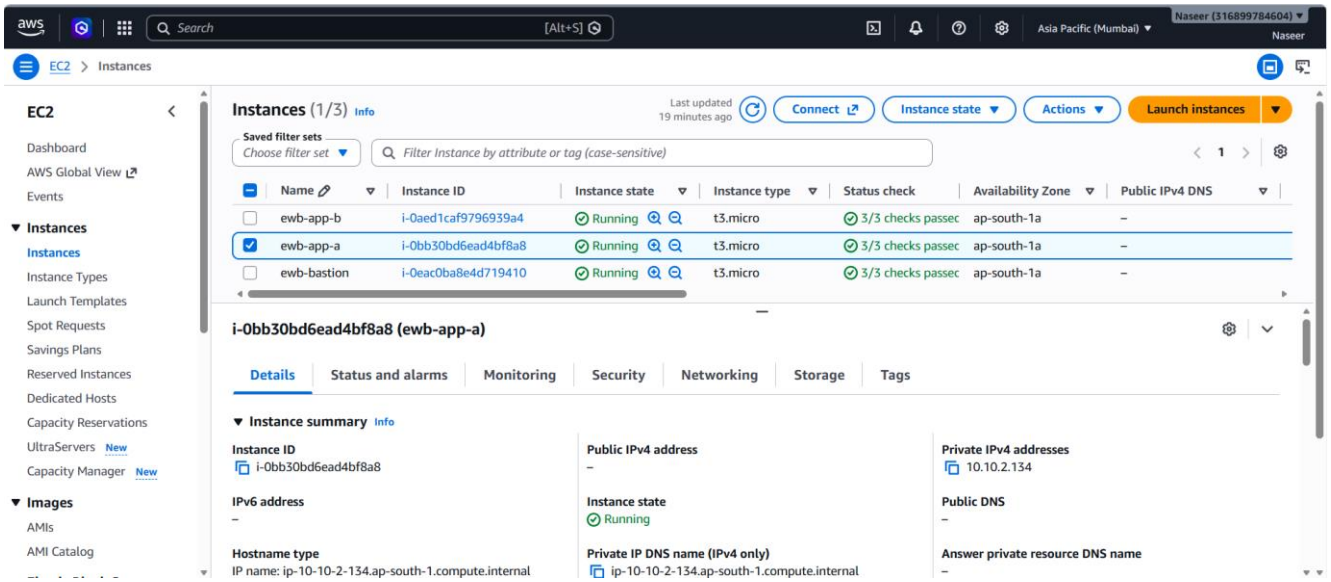
Subnet associations shown in screenshots

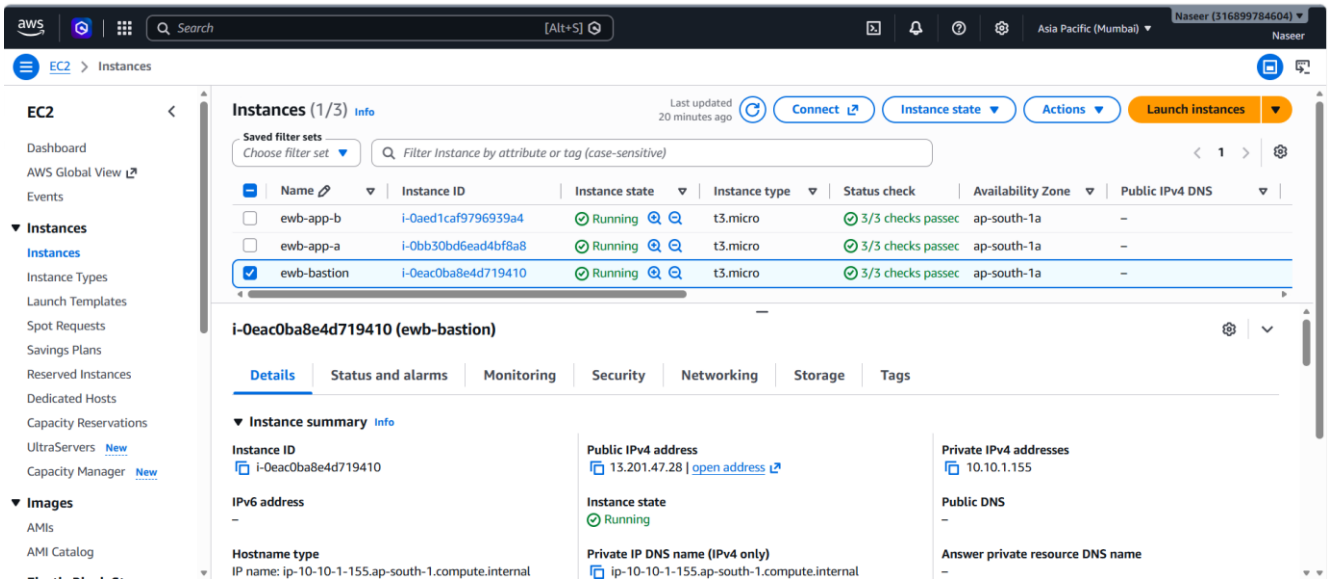
Route table	Subnet	Subnet ID	CIDR
ewb-rt-a-public	ewb-a-public	subnet-0e4d721a1e4fbe561	10.10.1.0/24
ewb-rt-a-private	ewb-a-private	subnet-0f748cc67c6e3c6f7	10.10.2.0/24
ewb-rt-b-private	ewb-b-private	subnet-0f04d830bbbf8fac9	10.20.1.0/24

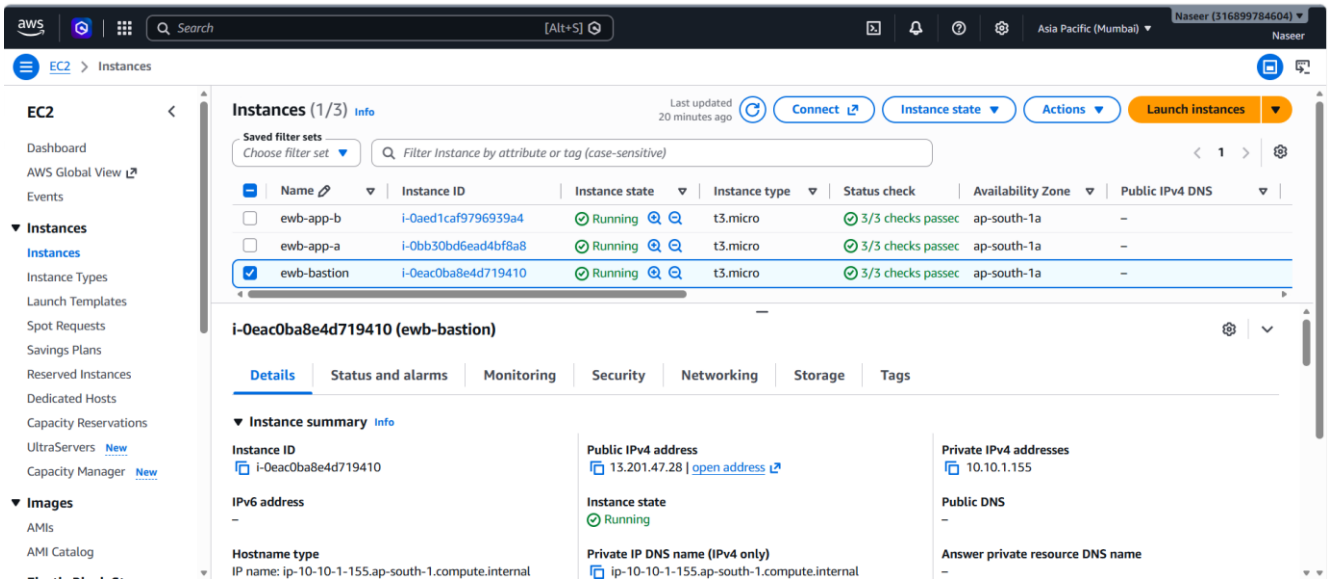
The public route table has an internet-gateway default route and a route to VPC B. Both private route tables contain the peer route and local route. The supplied association screenshots match the intended mapping.

5. Implementation and SSH Access

The terminal evidence shows successful TCP connectivity to the bastion on port 22 and SSH access to App-B through the bastion. The command `ssh ewb-app-b` connected to Amazon Linux 2023, and `hostname -I` returned `10.20.1.84`. Keep private keys local; never commit them to GitHub.






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ap-south-1a
i-0eac0ba8e4d719410 (ewb-bastion)
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Instance ID
i-0eac0ba8e4d719410
Public IPv4 address
13.201.47.28 | open address
Private IPv4 addresses
10.10.1.155
IPv6 address
Instance state
Running
Public DNS
Hostname type
IP name: ip-10-10-1-155.ap-south-1.compute.internal
Private IP DNS name (IPv4 only)
ip-10-10-1-155.ap-south-1.compute.internal
Answer private resource DNS name

6. Connectivity Testing and Results

Test	Observed result	Status
SSH to bastion TCP/22	Successful in supplied screenshot	Pass
Forward ping: 10.10.1.155 → 10.20.1.84	4 replies; 0% loss; average RTT ~0.206 ms	Pass
Reverse ping: 10.20.1.84 → 10.10.1.155	4 replies; 0% loss; average RTT ~0.162 ms	Pass
SSH to App-B through bastion	Successful; hostname -I returned 10.20.1.84	Pass
Ping from 10.10.1.52 to 10.20.1.0	4 transmitted, 0 received; 100% loss	Not controlled test

The failed ping targeted 10.20.1.0, the subnet network address, not App-B's host address 10.20.1.84. It is not proof of a controlled security-group failure.

7. Troubleshooting and Remaining Work

For failed cross-VPC communication, verify peering state, non-overlapping CIDRs, routes in both directions, subnet associations, security groups, network ACLs, host firewall, destination IP and SSH configuration.

Remaining items

■ Reconcile App-A instance ID, private IP and actual subnet association.

■ Verify current bastion public IP.

■ Capture actual security group inbound rules.

■ Complete controlled ICMP failure and recovery by removing only App-B's ICMP Echo Request rule from App-A's actual private IP /32, testing, restoring the rule and retesting.

■ Confirm both App-A and App-B have no public IPv4 address.

■ Capture cleanup evidence after saving project evidence.

8. Key Learning Outcomes

Practised VPC CIDR planning, VPC peering, route tables, subnet associations, security groups, bastion-based SSH access, ICMP testing, and troubleshooting network reachability.

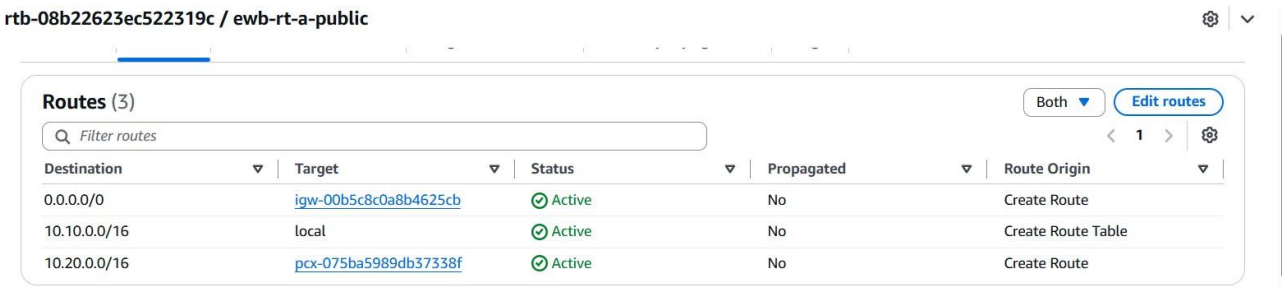
9. Conclusion

The supplied screenshots show expected active routes and subnet associations. Forward and reverse ping tests each returned four replies with 0% packet loss, and SSH access to App-B through the bastion was demonstrated. Controlled ICMP failure and recovery remains pending; the App-A address/subnet discrepancy must be resolved before final submission.

Appendix — Supplied Evidence Screenshots

Screenshots are included as supplied and are not independent access to the AWS account. The failed ping to 10.20.1.0 is labelled separately because it does not establish the controlled ICMP test.

A. Public route table



rtb-08b22623ec522319c / ewb-rt-a-public

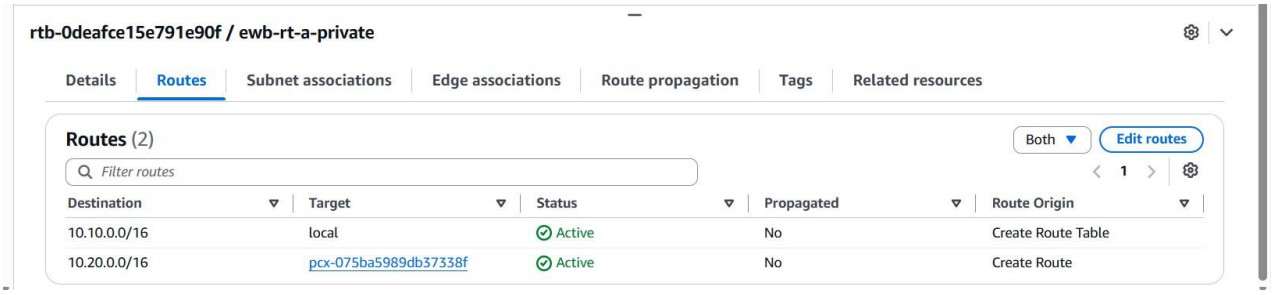
Routes (3)

Both | Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-00b5c8c0a8b4625cb	Active	No	Create Route
10.10.0.0/16	local	Active	No	Create Route Table
10.20.0.0/16	pcx-075ba5989db37338f	Active	No	Create Route

B. Private route table A



rtb-0deafce15e791e90f / ewb-rt-a-private

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags | Related resources

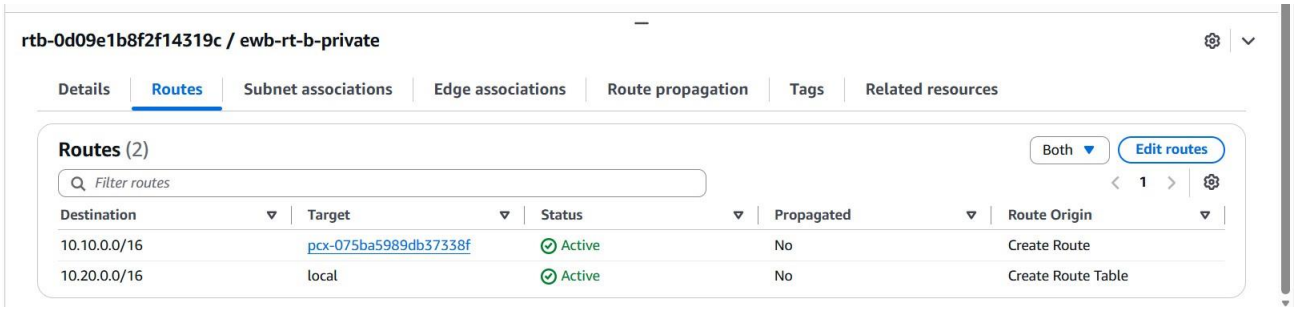
Routes (2)

Both | Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
10.10.0.0/16	local	Active	No	Create Route Table
10.20.0.0/16	pcx-075ba5989db37338f	Active	No	Create Route

C. Private route table B



rtb-0d09e1b8f2f14319c / ewb-rt-b-private

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags | Related resources

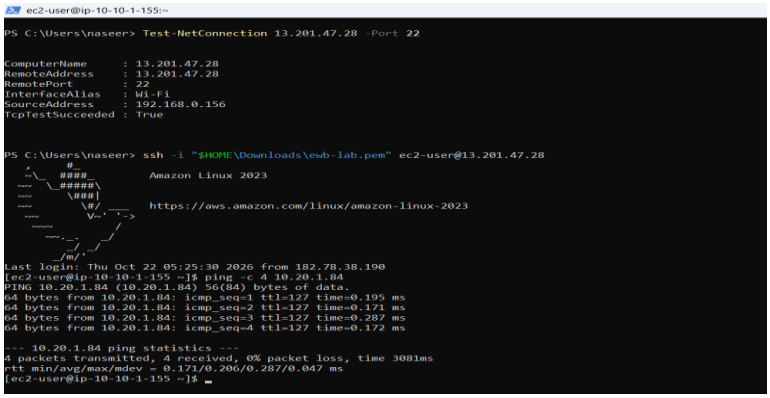
Routes (2)

Both | Edit routes

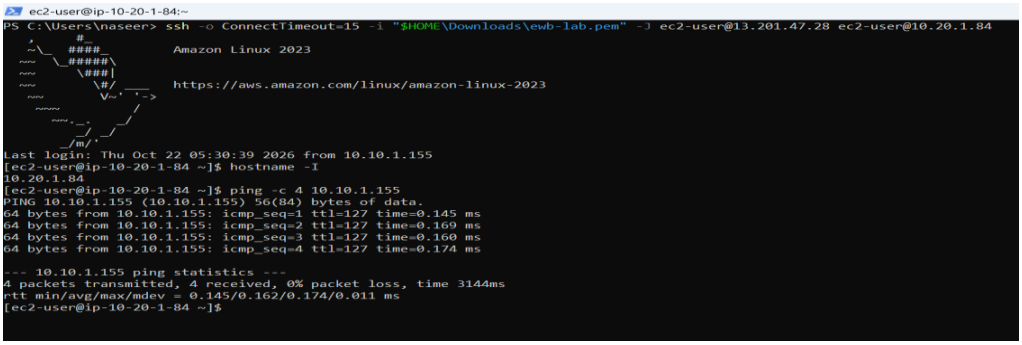
Filter routes

Destination	Target	Status	Propagated	Route Origin
10.10.0.0/16	pcx-075ba5989db37338f	Active	No	Create Route
10.20.0.0/16	local	Active	No	Create Route Table

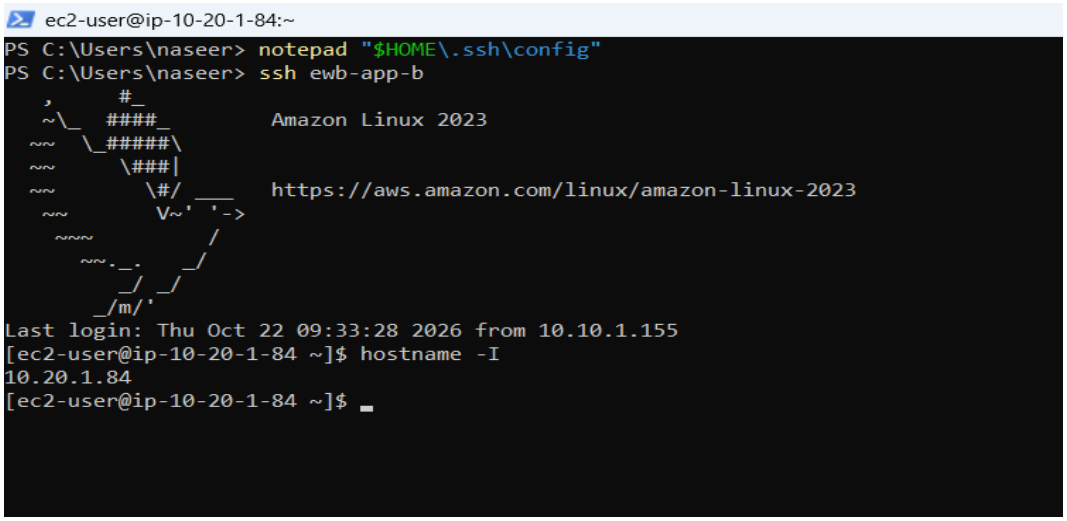
D.Forward ping



E.Reverse ping



F. SSH access to App-B



G. Failed ping to subnet address (not controlled tes

